

Josh Steckel

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EDUCATION

University of Delaware

Expected 2029

Bachelor of Engineering (B.E.) - Electrical Engineering | GPA: 3.99 | Dean's List

SKILLS

Core Skills: PCB Design, RF and Digital Signal Processing, FPGA and Digital Design, Embedded Systems, Systems Programming

Hardware & Test Equipment: Oscilloscope, Spectrum Analyzer, Vector Network Analyzer (VNA), Multimeter

Programming Languages: C, C++, Python, MATLAB, VHDL

Software & Tools: KiCad, Vivado, GNU Radio, SolidWorks

EXPERIENCE

Ground Station Team Lead

05/2026 – Present

Delaware Atmospheric Plasma Probe Experiment (DAPPEr)

- Led a 5-member sub team designing and upgrading ground station support equipment for a student-built LEO CubeSat mission
- Developed and tested channel models in MATLAB to determine optimal constant and adaptive downlink data rates for the UHF spacecraft communications link, achieving 9600 baud FSK downlink at 10 dB link margin
- Evaluated mission link budget and communications system performance for NASA's CubeSat Launch Initiative Pre-Procurement Readiness Review (PPRR) and FCC licensing

Crew Member - AMC Theatres, Inc., Allentown, PA

05/2024 – 08/2024

- Delivered guest services in a high-volume, fast-paced environment, operating concessions and maintaining a clean, organized, and safety-conscious space while collaborating with a large team

PROJECTS

Altoids Tin Meshtastic Node | *PCB Design, ESP32, LoRa, RF Design*

- Designed and assembled a 2-layer impedance-controlled RF PCB integrating an ESP32 MCU, UHF LoRa module, GPS module, user interface, and TFT LCD display
- Built and tested a LiPo battery-powered Meshtastic LoRa node housed in an Altoids tin, enabling decentralized, off-the-grid text communications and validating RF performance with lab test equipment

University Transit System Departure Board | *Embedded Systems, ESP32, FreeRTOS, Python, AWS*

- Designed a compact OLED arrival/departure board compatible with any ETA Transit-powered system, built on an automated headless web scraper (Selenium, Docker) hosted on AWS Lambda
- Implemented multi-core RTOS firmware on a low-power ESP32 client fetching pre-processed JSON data over secure HTTPS, cutting load time by 75% while running a horizontally scrolling marquee display

Fault Injection on PQC Algorithm Research | *Hardware Security, ChipWhisperer, ARM, Post-Quantum Cryptography*

- Contributing to VIP project team using ChipWhisperer boards to introduce instruction skips to ARM microcontrollers running Post-Quantum Cryptography (PQC) encryption algorithms (i.e. Kyber)
- Analyzing responses to tune glitch parameters to optimize fault injection windows

Power Side-Channel Attacks Research | *Machine Learning, CNNs, Signal Processing, Python*

- Trained and tested CNN models on 60+ hours of VR power consumption data to classify device activity from STFT frequency-spectrum images, achieving 97% classification accuracy and strengthening applied signal processing and data analysis skills

J-Pole Antenna Design and Installation | *Antenna Design, RF Engineering, Impedance Matching*

- Applied antenna theory to design and tune a 145 MHz VHF J-pole antenna, replacing the Amateur Radio Club's damaged repeater antenna
- Optimized gain and impedance matched antenna with coaxial cable to recover full repeater coverage
- Coordinated material costs as the club's treasurer and recorded purchases for reimbursement

ACTIVITIES, LEADERSHIP & CERTIFICATIONS

- IEEE — Member
- Amateur Radio Club — Treasurer
- FCC Amateur Radio License – Technician Class (KD3DKR)